



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CCS375 Web Technologies

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Servlet

- **Course Outcome:** CO3

- **Topic Learning Outcome:** 3a

- **Activity Chosen:** Crossword Puzzle

- **Justification:**

Crossword puzzle activity is an engaging and an interactive way to recall key terms and concepts. By answering servlet definitions, functions, and configurations-related puzzles, students improve their knowledge in the topic of servlet.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- The servlet concepts and the lifecycle methods were taught to the students. After that at the end of the class the activity was conducted for the duration of 10 minutes.
- The students were asked to form a team of 2 members and gave instructions and timing details.
- The puzzle sheet was given to the team. The students solved the puzzle based on the given clues. They were encouraged to discuss within teams for collaborative learning.
- After the allotted time gets completed, the answers were reviewed in the class by discussing the significance of each term and how it relates to servlet concepts.
- The students involvement in the activity is shown in Fig.1 and the sample answers given by the students is shown in Fig.2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO11	PO12	PSO1
CO3	1	3	3	3	3	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

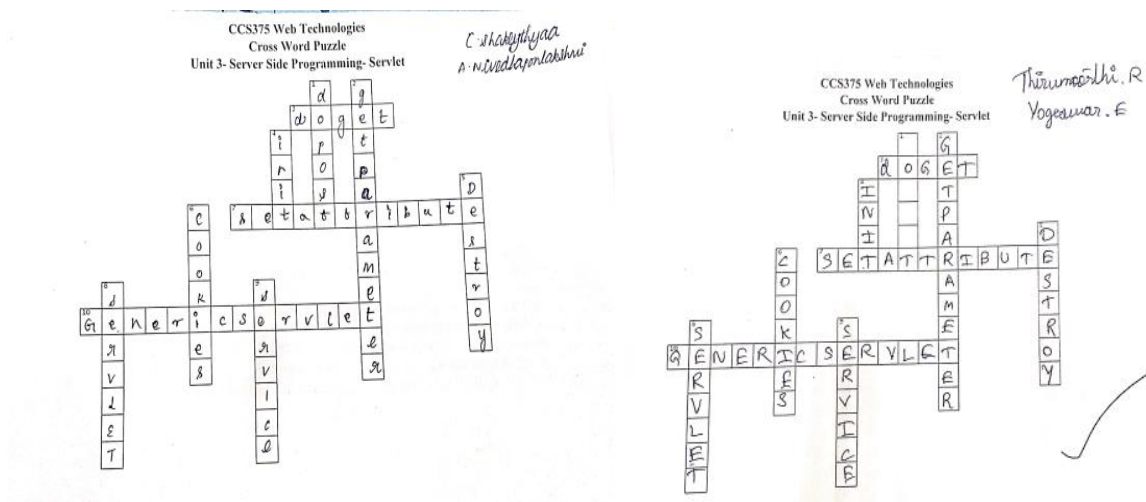
Innovative practice	PO1	PO2	PO9	PO10	PSO1
	1	2	1	1	3

Justification for correlation	The students apply the knowledge of server side programming for web application design	The students identify the modules to use in server side programming	The students combined as a team and solve problems	The students communicate their findings with their team	The server side programming concepts are needed in developing the web applications
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- Images / Screenshot of the practice:



Fig. 1: Students involvement



ACROSS

3. Which method is used to handle GET requests in a servlet?
7. Which method is used to add an attribute to the session in a servlet?
10. Which class is extended to create a servlet handle any type of protocol

DOWN

1. Write one non-lifecycle method of a servlet?
2. Which method of HttpServletRequest is used to retrieve a parameter sent by an HTML form?
4. Which lifecycle method is invoked when a servlet is first loaded into memory?
5. A method called to remove servlet from service and release any allocated resources.
6. A Way servlet maintain session information across multiple requests from the same client?
8. Which interface must be implemented by all servlets?
9. Which method is called by the servlet container to handle a client request?

Figure 2. Sample Crossword Puzzle Sheet

• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- All the students are very much interested in solving the crossword puzzle.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students familiarize themselves with servlet-specific terms and enhancing their technical language.
- The understanding of the concepts can be identified easily with the answers given by the students.

❖ ***Challenges faced in implementation:***

- It could be challenging for students with different levels of cognitive comprehension to engage equitably,
- Some students who are not familiar with some servlet-related terms could find it difficult to complete the challenge and need more help.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2_GS_CS8651_Crossword_Puzzles.pdf
- <https://puzzlemaker.discoveryeducation.com/criss-cross>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/RV-CWP-CS8601_MC_UIV.pdf